

Appendix:

Basic Workflow Syntax for “Shared Folder” and Dynamic Workflow

This document is intended to show the basic syntax used to create Shared Folder and Dynamic workflow rules and will go through an example PO Workflow document. This will cover a few specific workflow rules, which should give a feel for the types of rules that can be written but will not cover this aspect in extensive detail as the main goal of this document is only to illustrate basic syntax.

Note that this SOP focuses on static workflow rules and **not advanced dynamic workflow (ADW). For details on ADW rules visit the Help Center, accessed from “Search Help for a Solution” in a JAGGAER site. If logged in as a JAGGAER or Partner user, you will find additional documentation on workflow and other configuration topics.**

Static or “Shared Folder” and Dynamic Workflow

Steps

The Steps section shows the steps that customers can see on the approvals tab of a document. Each step is typically listed in order (although the order is not important in this section, it does help to be consistent with the intended order of workflow). The only necessary element is the Step name tag, which will contain the name of the step itself as show here:

```
<Step name="Missing Distribution" />
<Step name="Missing Distribution II" />
```

It is also possible to include help text with any workflow step that can be used to explain it's intended purpose. This is done by using the "wfActivityHelp" attribute with the help text included within the AttrValue element:

```
<Step name="PO Distribution">
  <Attribute name="wfActivityHelp">
    <AttrValue lang="default">System step to distribute a Purchase Order.</AttrValue>
  </Attribute>
</Step>
```

Activities

The bulk of workflow is contained in the activities section. This is where the rules are defined that control if a document will route to a step or not.

All activities contain the same basic elements, each of which is explained in the "Glossary" below:

- **Activity name** - This is just what it says, the name of the activity. This does not have to match the name of the step it is tied to, but often does
- **ActivityType** - This element will display the type of activity and can be any of the following:
 - **Manual** - This indicates that the step is manually evaluated by an end user. When this activity is active it must be approved by an organization approver.
 - **Automated** - This indicates that the step will complete automatically when the step becomes active.
 - **Notification** - This indicates that a notification will be sent out to users listed in the associated "Explicit Folder"
- **StepRef name** - This indicates the step that the activity is tied to. When workflow comes to this step, the activity will be triggered as well.
- **FolderRef name** - This is the shared workflow folder that is used to determine which users are able to approve the step. This folder must exist under Shared Workflow Folders.
- **Rule** - This is the meat of the activity. It contains all of the rules and evaluations that determine if a document is going to stop at the associated step or not.
- **FolderSelectionRule** - This indicates the type of folder that will be utilized for approvals and can be one of the following values:
 - **ExplicitFolder** - For manual activities, this will cause the step to use a Shared Workflow Folder. This can also be used for automated activities, which will point to a robot instead of a Shared Folder.
 - **MultiElementDynamic** - For automated activities. This indicates that the rule utilizes ADW rules for workflow approvals.

"START" STEP

The activities section of the document begins with a dummy "Start" step. This is required as a way for SelectSite to know where workflow should start and is followed by the first true activity. This is also necessary in the case that 2 steps start workflow in parallel:

```
<Activity name="START">
  <ActivityType>start</ActivityType>
  <Rule />
  <FolderSelectionRule>Dummy</FolderSelectionRule>
</Activity>
```

EXAMPLE ACTIVITY 1

Here is an example of a basic workflow activity:

```
<Activity name="Missing Distribution">
  <ActivityType>manual</ActivityType>
  <StepRef name="Missing Distribution" />
  <FolderRef name="Missing Distribution" />
  <Rule><![CDATA[ Document.HasInvalidDistribution() == true ]]></Rule>
  <FolderSelectionRule>ExplicitFolder</FolderSelectionRule>
</Activity>
```

Let's take a look at each line:

1. **<Activity name="Missing Distribution">** : The activity starts off with the name. This displays in SelectSite in the activity section.
2. **<ActivityType>manual</ActivityType>** : This activity is manual. This indicates that an approver from the organization must approve this step in order for the document to move on to the next step.
3. **<StepRef name="Missing Distribution" />** : This line shows the Workflow step that is tied to this activity. This activity will cause the step in this field to be triggered.
4. **<FolderRef name="Missing Distribution" />** : This line indicates the Shared Workflow folder that is tied to the activity, which defines the users that are able to approve the workflow step. In this case the Missing Distribution folder contains the users with approval powers.
5. **<Rule><![CDATA[Document.HasInvalidDistribution() == true]]></Rule>** : This section defines the rule for the activity. Rules will always be contained within the **<![CDATA[]>** element.
 - a. In this case the rule is **Document.HasInvalidDistribution() == true**.
 - i. The prefix of this rule is **Document**. This means that the rule evaluation is for the entire document. Alternately, if a rule contains the prefix "DocLine." the rule evaluates individual line items on the document.
 - ii. The suffix of this rule defines the "object" that is being triggered for the evaluation, in this case **HasInvalidDistribution()**. This object checks to see if the document is missing distribution information or if it is somehow invalid.
 - iii. The last part of this rule **== true** says that if the **Document.HasInvalidDistribution()** evaluation is true that the document will stop at the associated step. If this rule read **== false** instead, then the document would not stop here.
 - NOTE: Use of double equals signs (==) is required for this type of evaluation. A single equal sign (=) will not work.

6. **<FolderSelectionRule>ExplicitFolder</FolderSelectionRule>** : This line indicates the folder that the type of folder being used for approvals is an Explicit Folder. In other words, this rule uses a Shared Workflow folder, which is populated with approvers that have the ability to advance the document to the next step in workflow.
7. **</Activity>** : Each activity must end with this element. This indicates that the activity stops here.

EXAMPLE ACTIVITY 2

Next let's look at an example that uses an Advanced Dynamic Workflow rule evaluation (only new concepts will be explained in detail):

```
<Activity name="Director Approval">
  <ActivityType>manual</ActivityType>
  <StepRef name="Director Approval" />
  <Rule><![CDATA[ Document.RequiresRuleGroupEvaluation("Director Approval") == true ]]></Rule>
  <FolderSelectionRule>MultiElementDynamic</FolderSelectionRule>
  <Attribute name="RuleGroupInternalName">
    <AttrValue lang="default">Director Approval</AttrValue>
  </Attribute>
</Activity>
```

This rule shows some new concepts:

1. **<Rule><![CDATA[Document.RequiresRuleGroupEvaluation("Director Approval") == true]]></Rule>**: This line shows an ADW rule.
 - a. **Document.** indicates that this evaluates the entire document.
 - b. **RequiresRuleGroupEvaluation** shows that an ADW rule is being evaluated. In this case the rule being evaluated is **Director Approval**. This rule can be found in the Advanced Dynamic Workflow section of workflow configuration. There many rules can be configured under ADW. A document stops here if any one of the ADW rules in the group are true.
2. **<Attribute name="RuleGroupInternalName">**: The attribute name line will display in SelectSite and makes it easy to see which ADW rule group is evaluated in this activity.
 - a. **<AttrValue lang="default">Director Approval</AttrValue>** : This shows the rule group name itself.

EXAMPLE ACTIVITY 3

The next activity shows an automated step that requires multiple rule evaluations:

```
<Activity name="PO Distribution">
  <ActivityType>automated</ActivityType>
  <StepRef name="PO Distribution" />
  <FolderRef name="Distribute PO" />
  <Rule><![CDATA[ !(Document.CustomFieldValues("Manual Purchase Order") in {"true"}) &&
!(Document.CustomFieldValues("Order Category") in {"2 - Confirming - Do Not Distribute"}) ]]></Rule>
  <FolderSelectionRule>ExplicitFolder</FolderSelectionRule>
</Activity>
```

This activity covers a couple new ideas:

1. **<ActivityType>automated</ActivityType>** : This rule will not require a user to approve the step. This means that the step will complete automatically.
2. **<FolderRef name="Distribute PO" />** : This line may look familiar from the previous activities, but since this step is automated it works a little bit differently. Automated activities do not use a Shared Workflow step like manual activities. Instead automated activities take advantage of workflow robots. Robots are what allow automated steps to function without user approval. You can see what robot is being used for this folder under the Workflow Folder Import/Export tab in workflow setup.
3. **<Rule><![CDATA[!(Document.CustomFieldValues("Manual Purchase Order") in {"true"}) && !(Document.CustomFieldValues("Order Category") in {"2 - Confirming - Do Not Distribute"})]]></Rule>**:

This rule shows some interesting concepts:

1. **!(Document.CustomFieldValues("Manual Purchase Order") in {"true"})**
 - a. This shows that the rule is looking at a Custom Field value at the document level.
 - b. The rule is looking to see if the value of the **Manual Purchase Order** Custom Field has a value of **true**.
 - c. Notice the **!** in front of this rule. If you see **!** at the beginning of any rule, then a document will only stop if that rule IS NOT true. In this case, as long as the **Manual Purchase Order** is not **true**, the document will stop here.
2. **&&** - When you see two "and" symbols like this (&&) that means that the rule following the preceding rule MUST also be true. This allows customers to require several criteria in order for a document to stop. Many && can be used in a given activity.

EXAMPLE ACTIVITY 4

Finally, let's look at an activity that stops documents based on different possible rules:

```
<Activity name="EH&S Notification">
  <ActivityType>notification</ActivityType>
  <FolderRef name="EH&S Notification" />
  <Rule><![CDATA[ [Product.Radioactive = true || Product.RadioactiveMinor = true || Product.HazMat =
true || Product.Controlled = true || Product.Toxin = true || Product.SelectAgent = true] ]]></Rule>
  <FolderSelectionRule>ExplicitFolder</FolderSelectionRule>
</Activity>
```

Let's take a look at the rule itself:

1. **<Rule><![CDATA[[Product.Radioactive = true || Product.RadioactiveMinor = true || Product.HazMat = true || Product.Controlled = true || Product.Toxin = true || Product.SelectAgent = true]]]></Rule>**:
 - a. Notice the || between each rule evaluation, this represents an OR. This means that a document will stop if any one of these rules is true. Unlike the && found in the above example, all of these rules do not have to be true for the rule to be triggered. This allows an activity to be triggered based on separate and distinct criteria.
 - b. The rules in this activity are looking at the products contained in the document. This is indicated by the **Product.** prefix. Each product here contains different product attribute flags. This allows customers to stop documents that contain any product they want that contains any one of these attribute flags.

TIPS AND TRICKS



Workflow Language

"==" means "is equal to"

```
Document.RequiresFormTypeApproval() == true
```

"!=" means "not equal to"

"&&" means "and"

```
Document.RequiresRuleGroupEvaluation("AP Review") == true] &&
[Document.RequiresRuleGroupEvaluation("Omit from all but form approval") == false]
```

"||" means "or"

```
[[ ([Product.Radioactive = true] || [Product.RadioactiveMinor = true] || [Product.Hazardous =
true] || [Product.Controlled = true] || [Product.SelectAgent = true]
```